

LOOTING DETECTION USING SATELLITE IMAGES

UNDERGRADUATE RESEARCH PROPOSAL SUBMITTED
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1. INTRODUCTION

1.1 Motivation and Overview

Archaeological looting is one of the most serious threats to cultural heritage. It involves the illegal digging and removal of artifacts from archaeological sites, causing permanent damage to the archaeological context and reducing the scientific and historical value of the site [2], [4]. Looting often occurs in remote or insecure regions where regular field inspection is difficult, costly, or unsafe. Therefore, researchers and heritage authorities need efficient large-scale monitoring methods to identify damaged sites early.

Satellite remote sensing provides a practical solution for this problem. Satellite images can cover large geographical areas and provide repeated observations over time, making them useful for identifying looting-related surface disturbances such as pits, exposed soil, and texture changes [3], [4]. Previous studies have shown that remote sensing can support archaeological site monitoring in different regions such as Iraq and Jordan [2], [3].

In real-world conditions, long temporal sequences may not always be available because of cloud cover, missing acquisitions, or limited historical archives [4]. Therefore, there is a need to develop looting detection methods that remain effective even when only limited satellite image time-series data are available.

1.2 Aims and Objectives

Aim:

To develop and evaluate a machine learning and deep learning based system for detecting archaeological looting using satellite imagery under limited time-series conditions.

Objectives:

- To create limited time-series settings by restricting each sample to a small number of frames.
- To implement and compare baseline models suitable for short image sequences.
- To develop an improved short-series detection approach using change-aware features and lightweight temporal fusion.
- To evaluate the system using suitable performance metrics such as Precision, Recall, F1-score, AUC(Area Under the Curve), and IoU (Intersection over Union) where applicable.
- To identify the main factors affecting performance under limited temporal data.

1.3 Research Scope

This research focuses on detecting archaeological looting using satellite image time series under limited temporal coverage, developing a reliable and deployable method that performs effectively with few frames and enhances heritage monitoring for faster risk assessment and site protection.

2. LITERATURE REVIEW AND RESEARCH GAP

2.1 Literature Review

Remote sensing has become an important tool for monitoring archaeological sites and detecting threats such as looting, conflict damage, and land-use changes [4]. Optical imagery has been widely used to identify looting-related surface changes through multi-temporal analysis. For example, Agapiou used Earth observation imagery to detect looting activity at Apamea in Syria, showing that temporal image comparison can reveal site disturbances [10]. Similarly, Mahmoud et al. proposed a machine learning-based automated change detection tool for monitoring archaeological threats, achieving an F1-score of 0.82 and demonstrating the usefulness of scalable remote sensing workflows [8].

SAR-based methods have also been explored for looting detection. SAR imagery has advantages such as cloud-penetrating capability and sensitivity to surface roughness changes. El Hajj used interferometric SAR coherence and machine learning for looting detection, while Boyoğlu et al. studied the detectability of looting pits using SAR simulation [1], [7]. These

works show that SAR can contribute valuable information, although performance depends on environmental and acquisition conditions.

At a broader monitoring level, several studies have demonstrated the practical use of remote sensing for archaeological protection. Laugier et al. integrated remote sensing monitoring into heritage protection activities in the Kurdistan Region of Iraq, while Kopij et al. analyzed looting activity over time in Southern Jordan [2], [3]. In addition, regional-scale mapping of archaeological mounds and looting has been supported using high-resolution DEM and satellite imagery [6].

More recently, deep learning methods have improved looting detection from satellite image time series. Vincent et al. proposed a state-of-the-art deep learning approach and introduced the DAFA-LS dataset as a benchmark for looting detection [9]. Their work demonstrated that time-series modeling can effectively distinguish real looting disturbances from seasonal and background variations.

2.2 Research Gap

Dependence on Long Time-Series Data: Existing state-of-the-art methods for archaeological looting detection rely on long satellite image time series [9]. However, such long-term temporal data are not always available for many archaeological sites due to temporal gaps, cloud cover, and inconsistent archives [4]. Therefore, there is a need for detection approaches that work effectively with limited satellite image time-series data.

3. METHODOLOGY

The proposed research follows a multi-stage framework that combines dataset preparation, short-series modeling, and performance evaluation to detect archaeological looting under limited time-series conditions.

3.1 Data Acquisition and Pre-processing

The main dataset used in this research is DAFA-LS, introduced by Vincent et al. [9]. This dataset contains satellite image time series for looted and non-looted archaeological sites.

The preprocessing stage includes:

- verifying temporal order and metadata,
- standardizing image sizes and resolutions,
- applying normalization,
- handling missing or corrupted frames, and
- preparing fixed training, validation, and testing splits.

3.2 Limited Time-Series Setup

To address the research gap, limited time-series versions of the dataset will be created by restricting each sample to a small number of frames. Example settings include $K = 1, 2, 3, 5$, and 8 .

Different frame selection strategies will be tested, such as:

- recent-window selection,
- uniform sampling,
- random sampling, and
- paired early/late frame selection.

Missing-frame simulation will also be applied by randomly dropping frames to mimic real-world gaps caused by cloud cover or irregular image acquisition [9].

3.3 Baseline Models

Several baseline model families will be implemented for comparison:

1. Single-frame baseline – image-only CNN using one frame.
2. Two-frame change baseline – early and late frame comparison using stacked or difference-based inputs.
3. Short temporal pooling baseline – frame-wise feature extraction followed by mean or max pooling.
4. Light temporal fusion baseline – frame encoding followed by a lightweight temporal module such as GRU or attention pooling.

3.4 Proposed Method

The proposed method is designed for strong detection performance under limited temporal data. It combines change-aware feature representation with lightweight temporal fusion.

- Change-aware representation: feature differences between reference and target frames will be used to highlight disturbance signals.
- Light temporal fusion: compact temporal modeling will be applied to combine short image sequences efficiently.
- Training-time frame drop: random frame removal during training will improve robustness to missing observations.

3.5 Evaluation Strategy

The models will be evaluated using suitable metrics depending on the output type of the task. These include:

- Precision, Recall, F1-score, and ROC-AUC for classification,
- IoU, Dice, or mAP (mean Average Precision) for localization or segmentation tasks.

Performance will be analyzed across different K values and frame selection strategies. Additional error analysis will be carried out to study the effects of seasonal changes, subtle disturbances, and complex backgrounds.

4. DATASET / ARCHITECTURE

4.1 Dataset Description

The main dataset used in this research is **DAFA-LS**, introduced by Vincent et al [9]. The dataset contains approximately 2,000 satellite image time series covering both looted and non-looted archaeological sites across diverse geographical regions. Each time series provides multi-temporal observations, capturing changes in surface conditions over time. Limited time-series samples will be created by restricting each site to a fixed number of frames, allowing systematic testing of how detection performance varies with reduced temporal information. This setup enables evaluation of the robustness of different detection models under practical conditions where only a few observations may be available.

4.2 System Architecture

The system is designed as an integrated framework for looting detection under short time-series conditions. It combines preprocessing, feature extraction, change-aware representation, temporal fusion, and final prediction in a unified pipeline. Preprocessing handles image normalization, resizing, and missing-frame handling. Feature extraction uses CNN-based encoders to capture spatial patterns, while change-aware representation highlights temporal disturbances. Temporal fusion integrates information across the limited frames, enabling robust detection, and the final prediction identifies looted or non-looted sites efficiently.

5. TIMELINE

Table 1: Research Timeline for Looting Detection Study

	Semester 6					Semester 7					Semester 8	
Weeks	1-3	4-6	7-9	10-12	13-15	1-3	4-6	7-9	10-12	13-15	1-3	4-8
Literature Review												
Bibliography												
Research Proposal making												
Data Collection & preprocessing												
Model implementation												
Model Evaluation												
Research Thesis Writing												
Research paper writing												

6. REFERENCES

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